Continuous Random Variables

6.1 Probability density functions

Definition of Probability Density Function (PDF)

A.K.A density function, a conutinuous version of the PMF, and this function should always output a non-negative real value

The probaility is $P(X \in A) = \int_A f(x)dx$

(a) $F(t) = \int_{-\infty}^{t} f(x)dx$

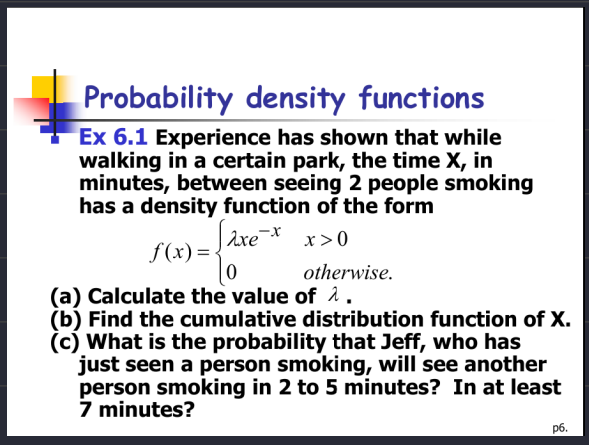
(b) $\int_{-\infty}^{\infty} f(x)dx = 1$

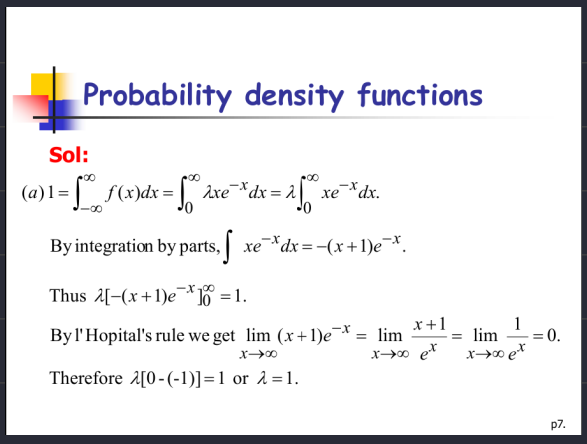
(c) If f is continuous, then F'(x) = f(x), 即使f不連續，在f連續的部分 F'(x) = f(x)

(d) For real number $a \le b$, $P(a \le X \le b) = \int_a^b f(t)dt$

(e) $P(a < X < b) = P(a \le X < b) = P(a < X \le b) = P(a \le X \le b) = \int_a^b f(t)dt$

表示在連續分布中 P(x) = 0

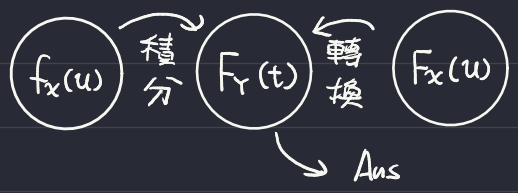




6.2 Density function of a function of a random variable

TLDR：給定一個隨機變數 X 與一個轉換函數 Y=h(X)，問你 F(Y), f(Y)

題目顆秒：



Ex 6.3 **Let X be a continuous random variable with the probability density function**

$$f(x)=\begin{cases}2/x^2 & if\ \ 1<x<2\\ 0 & elsewhere.\end{cases}$$

Find the distribution and density functions of $Y=X^2$.

1. $F_X(x)=\int_1^x 2x^{-2}dx = (-2x^{-1})\Big|_1^x = -2x^{-1}+2$

2. $F_Y(t) = P(Y\le t)$
   $= P(X^2\le t)$ ↘ 找 $X^2\le t$ 的上下限
   $= P(-\sqrt{t}\le X\le \sqrt{t})$ ↘ 注意原始函數範圍
   $= P(1\le X\le \sqrt{t})$
   $= F_X(\sqrt{t}) = -2t^{-\frac{1}{2}}+2$ $\quad 1\le t\le 4$ ← 找 $Y=X^2$ 上下限

3. $f_Y(t) = F_X'(\sqrt{t}) = t^{-\frac{3}{2}}$ $\quad 1\le t\le 4$, 0 otherwise.

6.3 Expectations and variances

Excepted value of Continuous R. V. Definition

A continuous R.V. with P.D.F f(x), has the mean/excepted value E(X):

$$E(X) = \int_{-\infty}^{\infty} x \times f(x)dx$$

TODO: p.28

Cauchy random variable

這邊只是證明積分後機率加總為 1，可以不用被證明過程

Theorem 6.2

For any continuous R.V. X with distribution function F and density function f,

$$E(X) = \int_0^{\infty} [1 - F(t)]dt - \int_0^{\infty} F(-t)dt$$

$$E(X) = \int_0^{\infty} P(X > t)dt - \int_0^{\infty} P(X \leq -t)dt$$

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^{0} xf(x)dx + \int_0^{\infty} xf(x)dx \\ &= -\int_{-\infty}^{0} (\int_0^{-x} dt) f(x)dx + \int_0^{\infty} (\int_0^{x} dt) f(x)dx \\ &= -\int_0^{\infty} (\int_{-\infty}^{-t} f(x)dx)dt + \int_0^{\infty} (\int_t^{\infty} f(x)dx)dt, \end{aligned}$$

where the last equality is obtained by changing the order of integratio n.

TL;DR: 在只有 F(t) 的情況下求 E(X)

也可直接將 F(X) 微分得到 f(x)

後直接對其作積分 (Def 1)

Theorem 6.3

Let X be a continuous R.V. with density function f(x);

then for any function h(x): R → R

$$E[h(x)] = \int_{-\infty}^{\infty} h(x) \cdot f(x)dx$$

且這邊的期望值也遵守線性三條

Definition of Varience

$$Var(x) = E(X^2) - (E(X))^2$$